

Morphing Camber Wings

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I. PROBLEM STATEMENT AND MOTIVATION

Conventional aircraft and UAV wing designs rely heavily on discrete control surfaces such as flaps, rudders, and ailerons to manipulate airflow and generate lift or torque. While effective at large scales, these mechanisms introduce geometric discontinuities that can cause flow separation and increased drag—especially detrimental at the low Reynolds numbers typical of small unmanned aerial vehicles (UAVs). Additionally, actuator-based control systems add mass, mechanical complexity, and failure modes that may be avoided through alternative methods of shape control.

In contrast, birds and other flying animals modulate their aerodynamic surfaces through smooth, continuous changes in wing shape. They can dynamically adjust camber, span, and twist in response to external conditions without discrete control surfaces. For example, swifts and falcons actively control wing curvature and twist during glide and maneuvering to optimize lift-to-drag ratio, delay stall, and respond to gusts [1]. This biologically-inspired approach to morphing suggests a path forward for low-mass, high-agility UAV designs.

The goal of this project is to explore avian-inspired wing morphing mechanisms, beginning with the simulation of camber changes in a 2D elastic beam. The beam represents a chordwise cross-section of a wing capable of passive or active shape change. This simplified model allows us to study how boundary actuation (e.g., changes in tip displacement or angle) can induce desired internal curvature using material compliance alone.

In the future, this framework can be extended to incorporate twist and eventually to 3D shell models, representing the full spanwise and chordwise morphing behavior of bird wings. By simulating such compliant systems under feedback control, we aim to better understand how simple mechanisms can replicate the performance advantages of avian flight without relying on complex actuation or high-fidelity aeroelastic modeling.

II. BACKGROUND AND LITERATURE REVIEW

Wing morphing has long been a topic of interest in both biological studies and aerospace engineering. Birds and bats achieve remarkable aerodynamic versatility through continuous geometric transformations of their wings—adjusting span, camber, and twist during flight to optimize for maneuverability, energy efficiency, and gust rejection. These biological insights have motivated research into compliant, actuator-minimal morphing structures in robotic and aerospace systems.

A. Biological Inspiration

Numerous studies have documented how birds employ non-rigid wing morphing for control and adaptation. Lentink et al. [1] showed that swifts actively control camber during gliding to improve aerodynamic efficiency. Rather than relying on discrete joints or actuators, birds use a combination of musculoskeletal structure and passive material deformation to enable such smooth transformations.

B. Engineering Implementations

Traditional morphing mechanisms in aerospace rely on embedded actuators and hinges [2], which introduce structural discontinuities and added weight. To address these issues, researchers have turned to compliant mechanisms and smart materials, such as shape memory alloys and piezoelectrics, to enable distributed deformation with fewer moving parts [3]. These approaches are particularly attractive for small UAVs operating at low Reynolds numbers, where drag from control surfaces is more detrimental.

Flexible beam and plate models have been used as simplified representations of morphing structures [4]. Bending-dominated deformations allow for camber-like effects, while torsional responses can simulate spanwise twist. However, these models often assume either fully passive deformation under aerodynamic loads or rely on high-fidelity simulations with embedded actuation.

C. Control Challenges

Controlling the deformation of compliant structures poses significant challenges. Recent studies have implemented feedback-based methods using PID or model predictive control to regulate shape in soft robotics and morphing airfoils [5], [6]. However, these often require full-state sensing or complex material modeling, limiting their practicality in lightweight UAV applications.

D. Gap and Contribution

Existing work either targets high-fidelity control of morphing structures using heavy or complex actuation, or focuses purely on passive aeroelastic deformation. This project proposes a middle-ground approach: a simple, planar elastic beam simulating chordwise camber controlled by minimal boundary actuation. By simulating this in 2D with real material properties and a simple PI controller, we aim to gain insight

into how bio-inspired compliance and shape control can be achieved with minimal sensing and actuation.

Unlike previous work that focuses on aerodynamic performance, our emphasis is on structural control and feedback in a reduced-order setting—serving as a foundation for future 3D shell simulations incorporating both camber and twist.

III. METHODS

A. Elastic Beam Model

We model the wing cross-section as a two-dimensional elastic beam discretized into a series of nodes and segments. Each segment possesses bending stiffness $EI(s)$, axial stiffness EA , and a corresponding curvature $\kappa(s)$. This formulation allows the beam to deform in response to internal actuation while maintaining computational efficiency. The model structure follows the reduced-order elastic rod formulation introduced in our proposal and is well suited for exploring shape changes driven by curvature actuation rather than external aerodynamic loading.

B. Curvature-Based Actuation

To simulate camber morphing, we prescribe a natural curvature distribution $\kappa_0(s)$ along the length of the beam. The beam adopts the configuration that minimizes its internal bending energy. This energy is expressed as

$$E_b = \frac{1}{2} \int EI(s) (\kappa(s) - \kappa_0(s))^2 ds, \quad (1)$$

which penalizes deviations of the actual curvature from the prescribed target curvature. Several curvature distributions have been implemented, including uniform, linear, exponential, and Gaussian profiles. Each of these profiles produces a characteristic deformation shape and allows us to explore a range of biologically inspired and engineering-relevant camber configurations.

C. Variable Stiffness

To more accurately reflect the heterogeneous structure of the biological wings, we incorporate spatially varying stiffness by assigning a stiffness distribution $EI(s)$ along the beam. For instance, assigning a larger bending stiffness near the leading edge and a smaller stiffness near the trailing edge promotes localized deformation where flexibility is desired. This feature reduces the actuation effort required to achieve a given camber shape and yields more realistic morphing behavior. The integration of stiffness gradients also enables the exploration of how passive structural compliance influences the achievable camber profiles.

D. Shape Reconstruction

Given a curvature distribution, the shape of the beam is reconstructed through a two-step geometric integration procedure. First, the curvature profile is integrated to obtain the angular orientation $\theta(s)$ along the beam. Next, the orientation is integrated to compute the Cartesian coordinates of each node, resulting in a complete reconstructed camber shape.

This approach avoids the computational complexity of solving nonlinear dynamic equilibrium equations and provides rapid evaluation of shape changes under different curvature and stiffness conditions.

IV. PROGRESS TO DATE AND PRELIMINARY RESULTS

Substantial progress has been made in developing the curvature-based camber morphing simulation. The primary milestone has been the successful implementation of prescribed natural curvature $\kappa_0(s)$ as the actuation mechanism for inducing beam deformation. This approach allows the beam to take on a wide range of biologically inspired shapes by defining curvature distributions along its span. To date, we have generated curvature fields using four families of functions—uniform, exponential, linear, and Gaussian—each producing distinct deformation characteristics. These curvature functions emulate different forms of structural actuation, enabling systematic exploration of their effects on camber.

In addition to curvature prescription, we have incorporated variable-stiffness capabilities by allowing the bending stiffness $EI(s)$ to change along the beam. This feature makes it possible to model wings in which the leading edge is significantly stiffer than the trailing edge. The resulting stiffness gradients substantially influence deformation magnitude and sensitivity, enabling greater deflection near the trailing edge without requiring larger actuation inputs. This behavior is consistent with biological feathers and compliant wing structures.

Figures 1 and 2 show examples of the prescribed curvature distributions for the uniform and exponential cases. The uniform curvature produces nearly circular-arc deformation, while the exponential curvature concentrates bending toward the trailing edge. The corresponding reconstructed beam shapes are shown in Figures 3 and 4. These results validate that the curvature-based formulation produces consistent and physically meaningful camber morphing behavior under both constant and spatially varying curvature fields.

In general, the preliminary results demonstrate that the simulation framework accurately captures the morphing of the static camber under a variety of inputs of curvature and stiffness gradients. This establishes a strong foundation for the next phase of the project, which will introduce dynamic actuation and transient analysis.

V. NEXT STEPS

The next phase of development focuses on expanding the simulation from static to dynamic morphology. We will implement a time-varying curvature transition that interpolates between two camber states. This will allow us to evaluate the transient response of the structure and assess factors such as the deformation rate, internal stresses, and actuation energy.

Once dynamic morphing is implemented, we will perform a series of parametric studies to examine the relationships between transition time, stiffness gradients, curvature magnitude, and actuation effort. These analyses will also include the evaluation of transient curvature distributions, internal moment

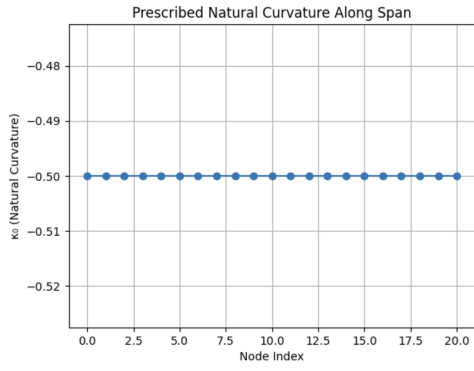


Fig. 1. Prescribed uniform natural curvature $\kappa_0(s)$ along the beam span.

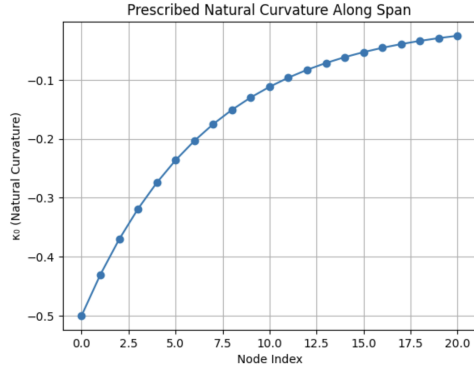


Fig. 2. Prescribed exponential natural curvature $\kappa_0(s)$, producing stronger trailing-edge bending.

evolution, and potential oscillatory behavior following rapid actuation.

We also plan to incorporate a proportional–integral (PI) controller to regulate morphed shapes by adjusting boundary actuation inputs. This will allow the system to track target camber angles or tip displacements dynamically. In later stages, approximate aerodynamic loading will be introduced to simulate more realistic wing behavior under airflow, using simplified distributed pressure or lift models.

VI. ANTICIPATED FINAL OUTCOMES

By the end of the project, we expect to deliver a comprehensive reduced-order simulation capable of reproducing both static and dynamic camber morphing. The final system will generate visualizations of time-dependent deformation, quantify internal forces and strain energy, and enable comparison between various curvature and stiffness profiles. The framework will also offer insight into how bio-inspired structural gradients and minimal actuation can produce efficient morphing suitable for lightweight UAV applications. Ultimately, this work will serve as a foundation for future studies involving coupled twist–camber morphing and full three-dimensional aeroelastic simulations.

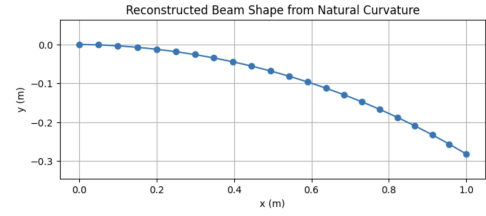


Fig. 3. Reconstructed beam shape resulting from uniform natural curvature. The deformation resembles a smooth circular-arc camber.

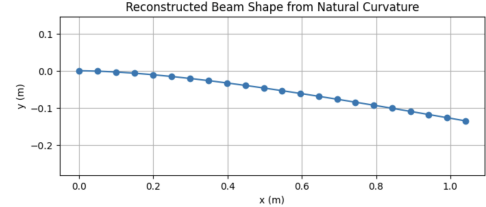


Fig. 4. Reconstructed beam shape resulting from exponential natural curvature. Curvature is concentrated toward the trailing edge, producing asymmetric deformation.

VII. CONCLUSION

The progress achieved thus far demonstrates that curvature-based actuation and variable stiffness provide a powerful and computationally efficient means of simulating continuous camber morphing. With dynamic morphing, controller integration, and aerodynamic loading planned for the next phase, the project is on track to deliver an effective and extensible platform for exploring bio-inspired morphing wing concepts.

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